

Transformers and embeddings

Dirk Wulff & Zak Hussain



MAX PLANCK INSTITUTE
FOR HUMAN DEVELOPMENT





John Rupert Firth
Linguist
1890-1960



**“You’ll know
a word by
the company
that it
keeps!”**

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Linguist
1890-1960

Word embedding

Latent semantic analysis

		Contexts										
Word	this region of the galaxy	Context 1	1									
			1									
			1									
			1									
			1									

"You're on your way, Kelvin. Good luck!" Moddard's voice sounded as close as before.

A wide slit opened at eye-level, and I could see the stars. The `_Prometheus_` was orbiting in the region of Alpha in Aquarius and I tried in vain to orient myself; a glittering dust filled my porthole. I could not recognize a single constellation; in this region of the galaxy the sky was unfamiliar to me. I waited for the moment when I would pass near the first distinct star, but I was unable to isolate any one of them. Their brightness was fading; they receded, merging into a vague, purplish glimmer, the sole indication of the distance I had already travelled. My body rigid, sealed in its pneumatic envelope, I was knifing through space with the impression of standing still in the void, my only distraction the steadily mounting heat.

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		Contexts											
Word		Context 1	Context 2										
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	region	1	1										
	of	1	1										
	the	1	2										
	galaxy	1	1										

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	region	1										
	of	1	1									
	the	1	1	1								
	galaxy	1	2	2								
	sky	1	1	1								
				1								

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		Contexts										
Word		Context 1	Context 2	Context 3	...	Context m						
	this	1										
	region	1	1									
	of	1	1	1		1						
	the	1	2	2		1						
	galaxy	1	1	1								
	sky			1								
	...											
	image					1						
	dial					1						
shivered						1						

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	region	1	1								
	of	1	1	1		1				1	
	the	1	2	2		1					
	galaxy	1	1	1					1		
	sky			1			2				1
	...				1						
	image					1					3
	dial					1			1		
shivered						1				1	
...											1
Word <i>N</i>									1		

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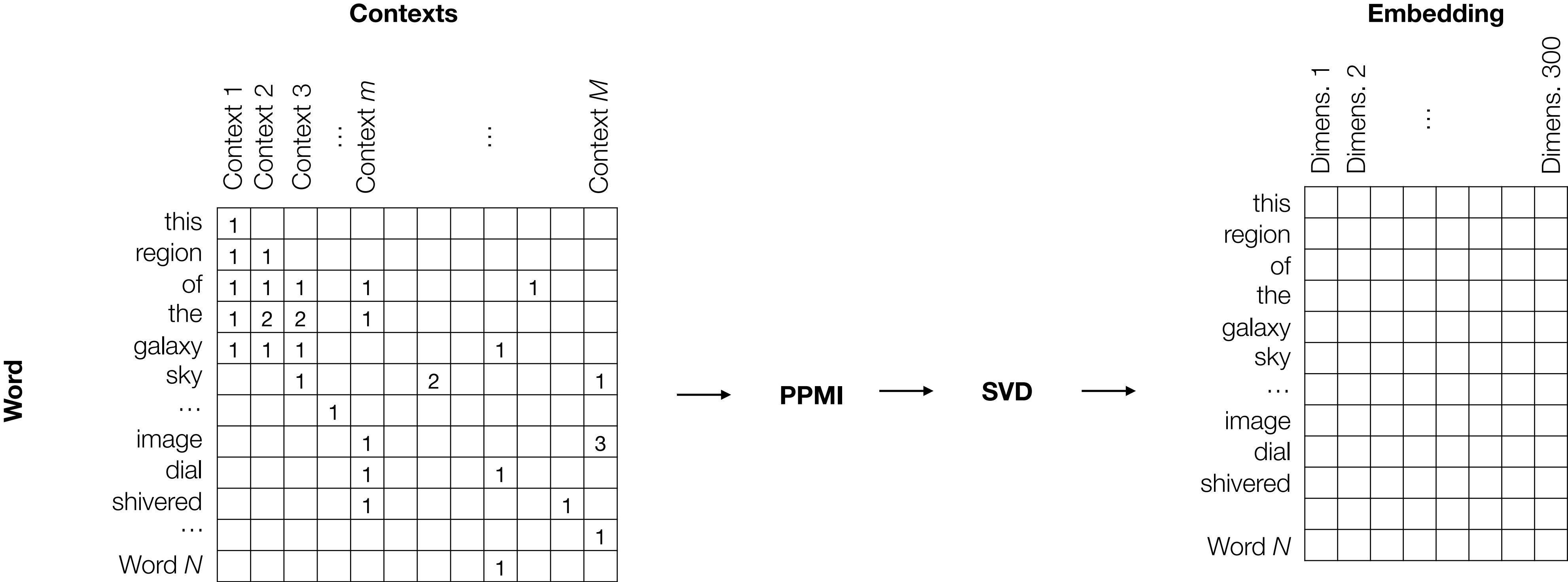
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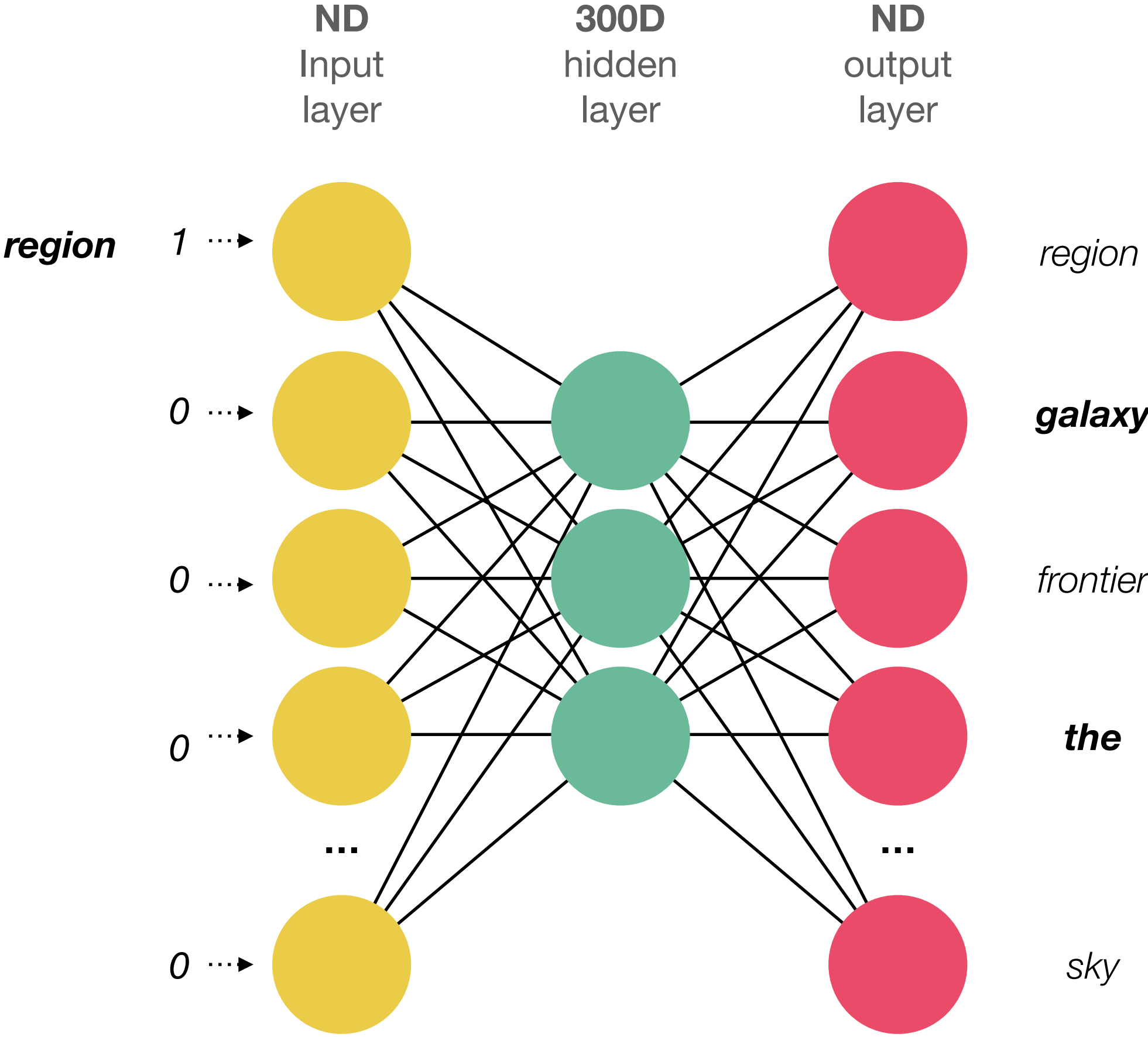
Word embedding

Latent semantic analysis



Word embedding

word2vec



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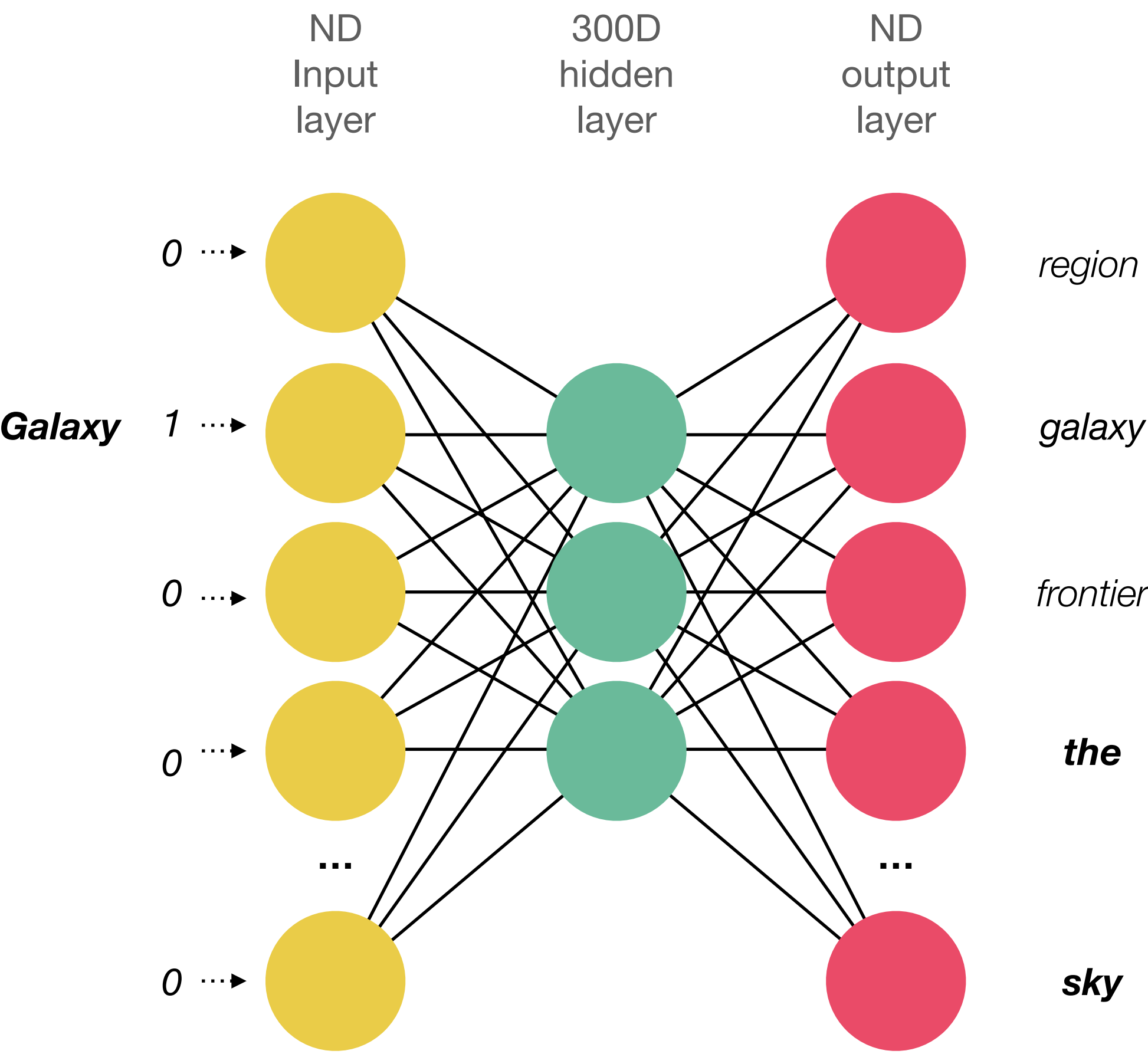
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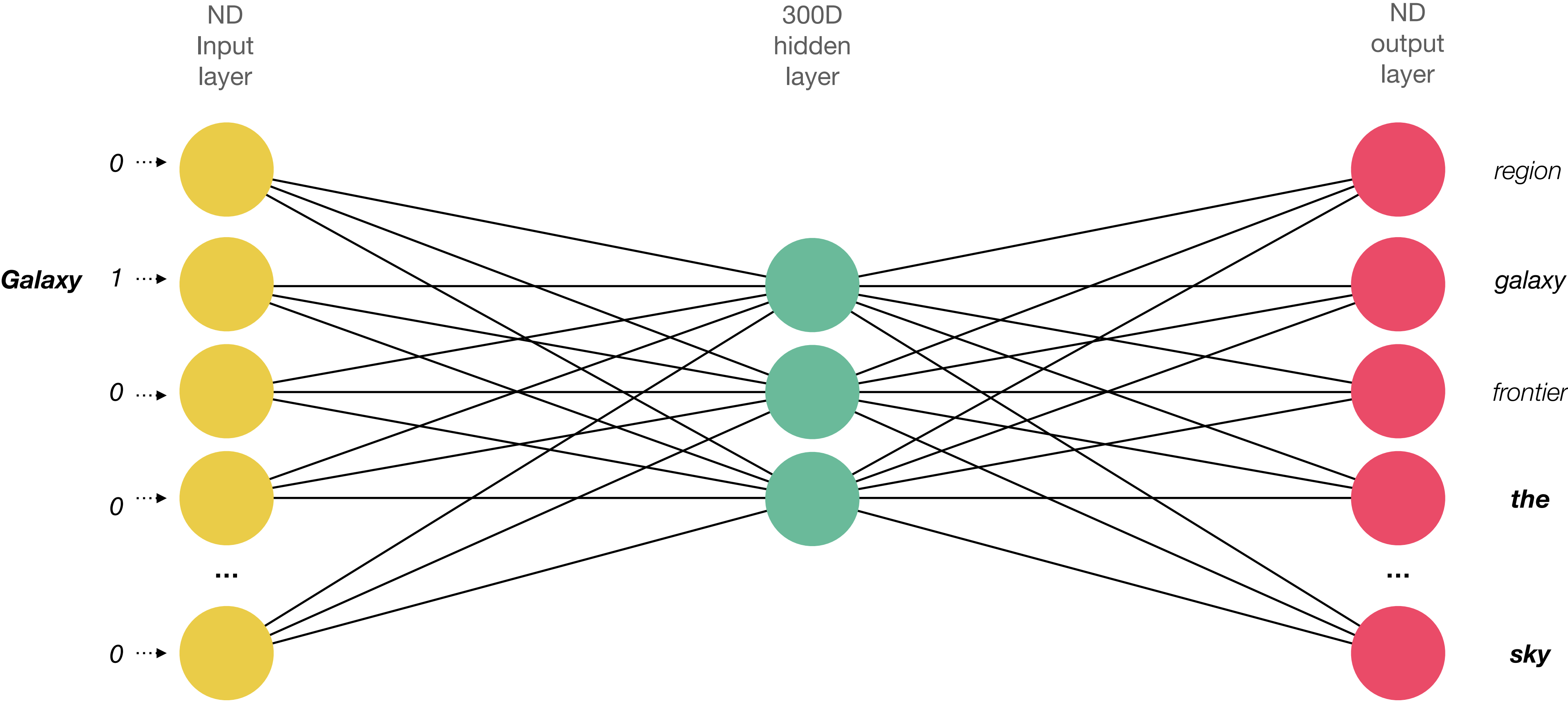
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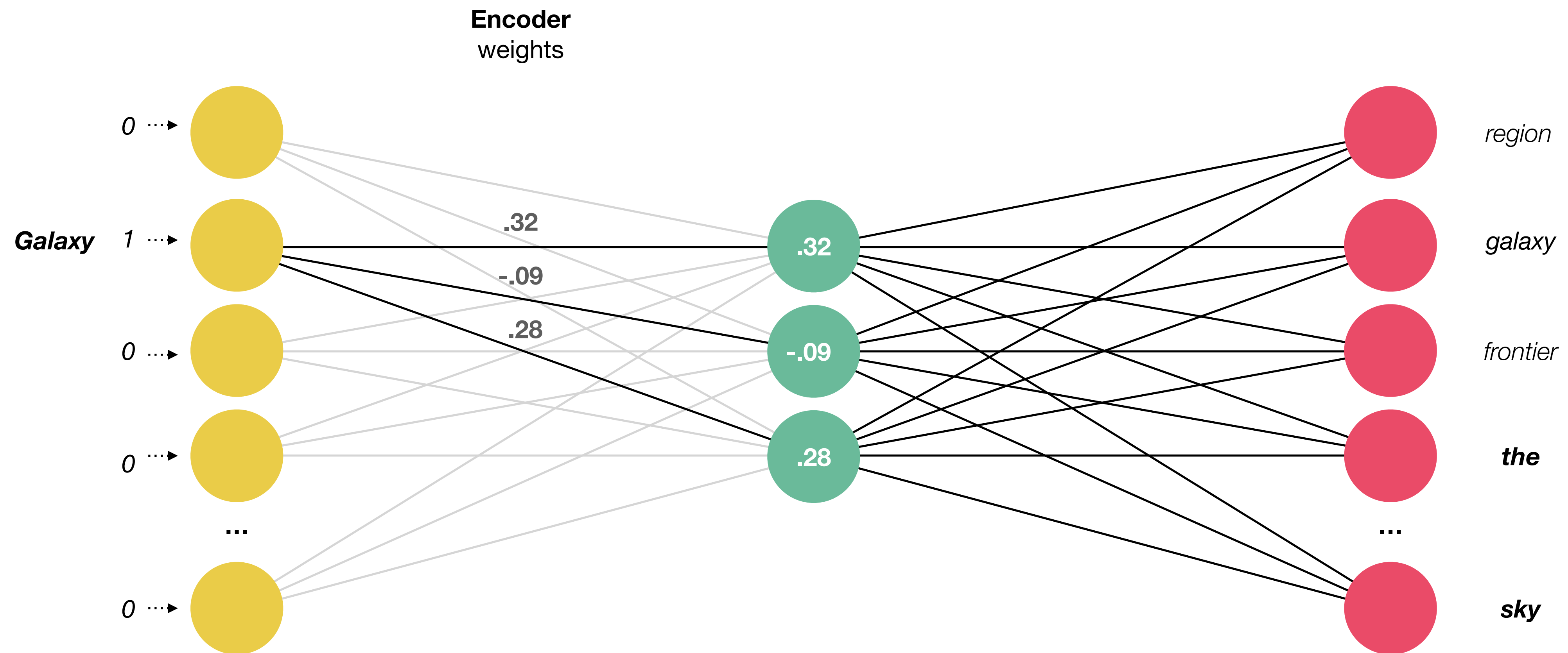
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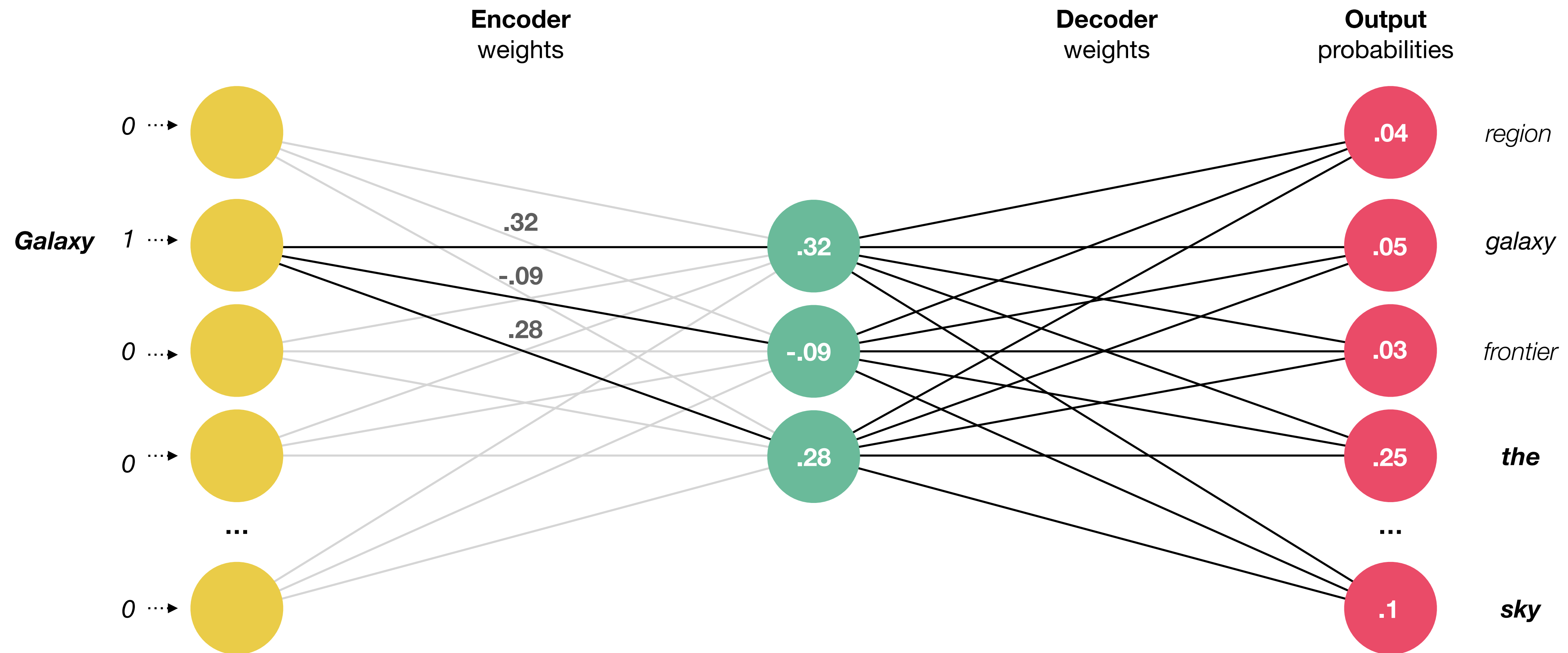
Word embedding

“A good language model encodes words with similar output similarly”



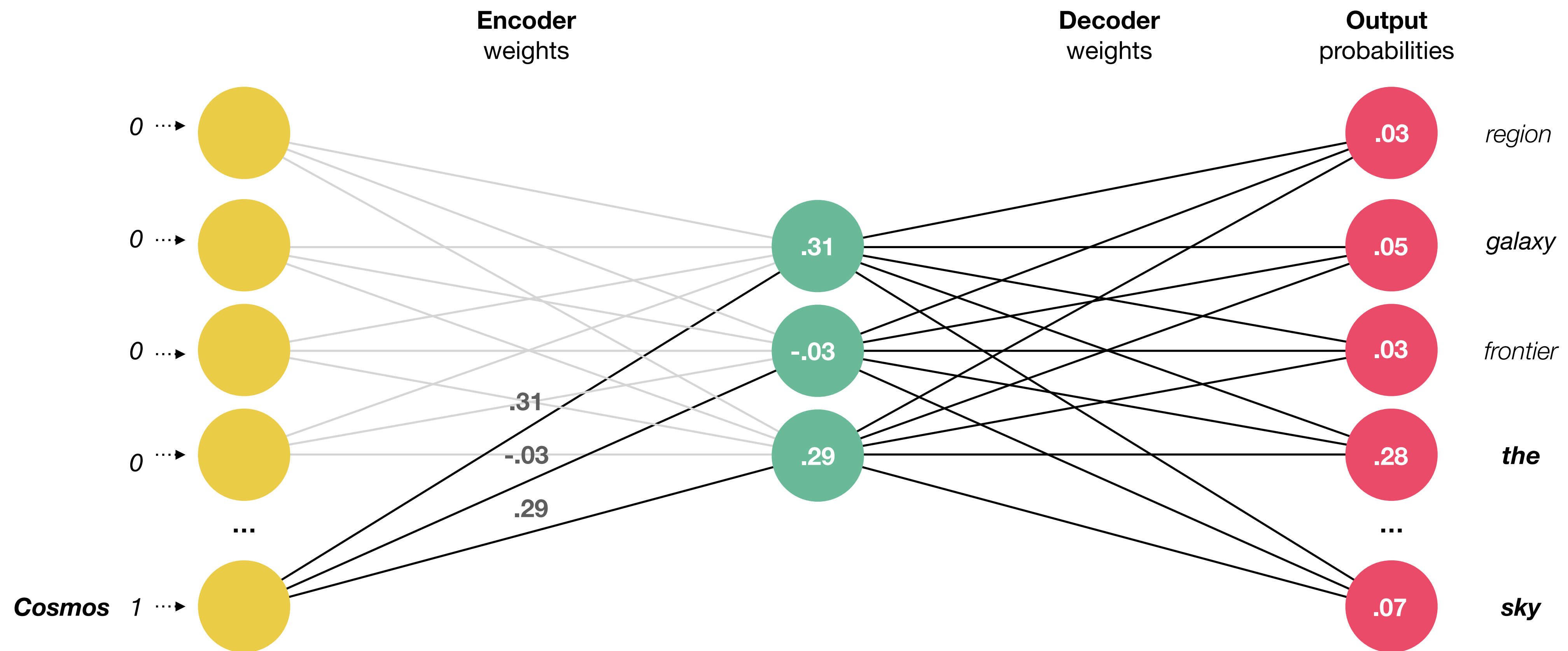
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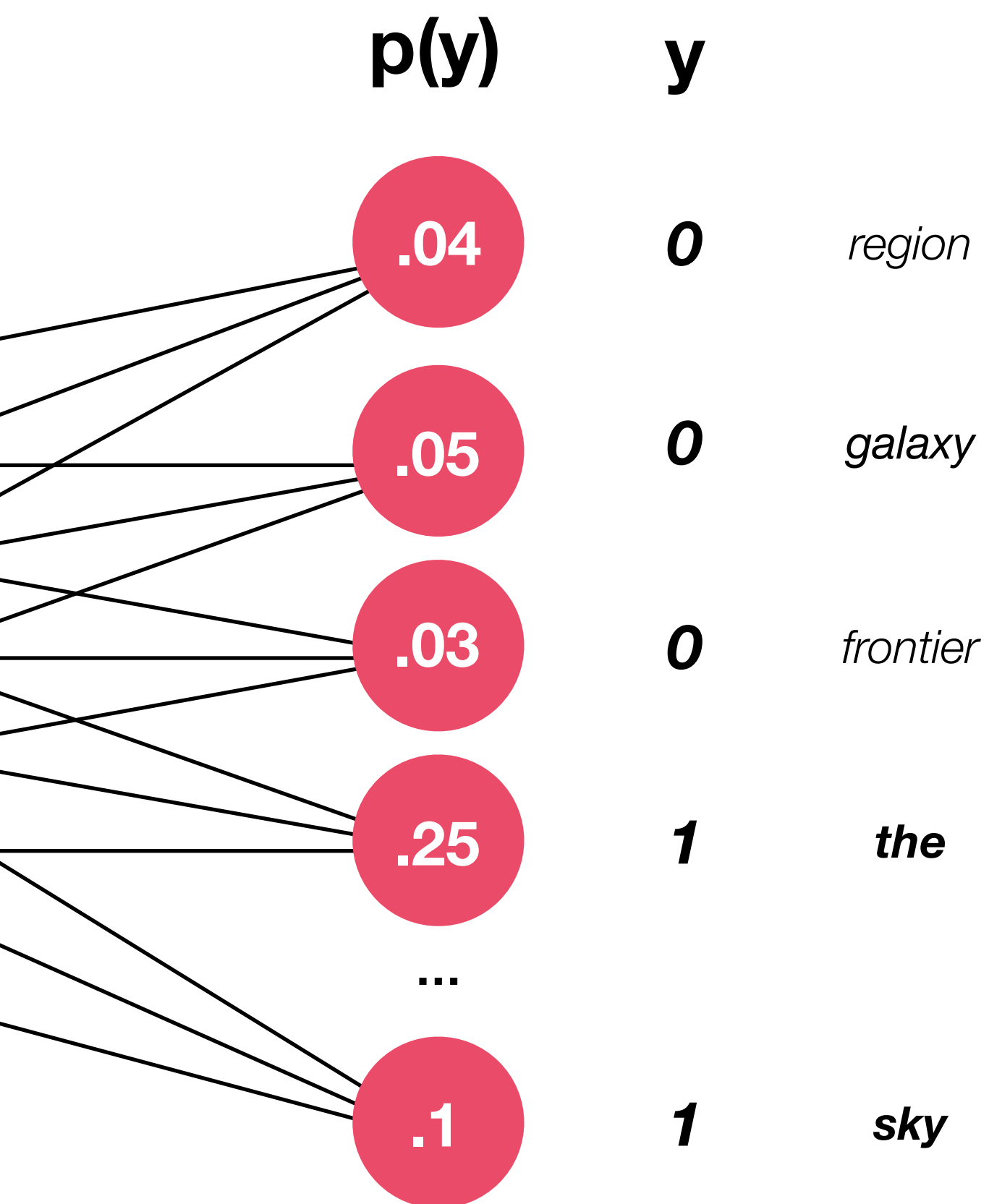
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“A good language model encodes words with similar context similarly”



Error signal

and gradient descent



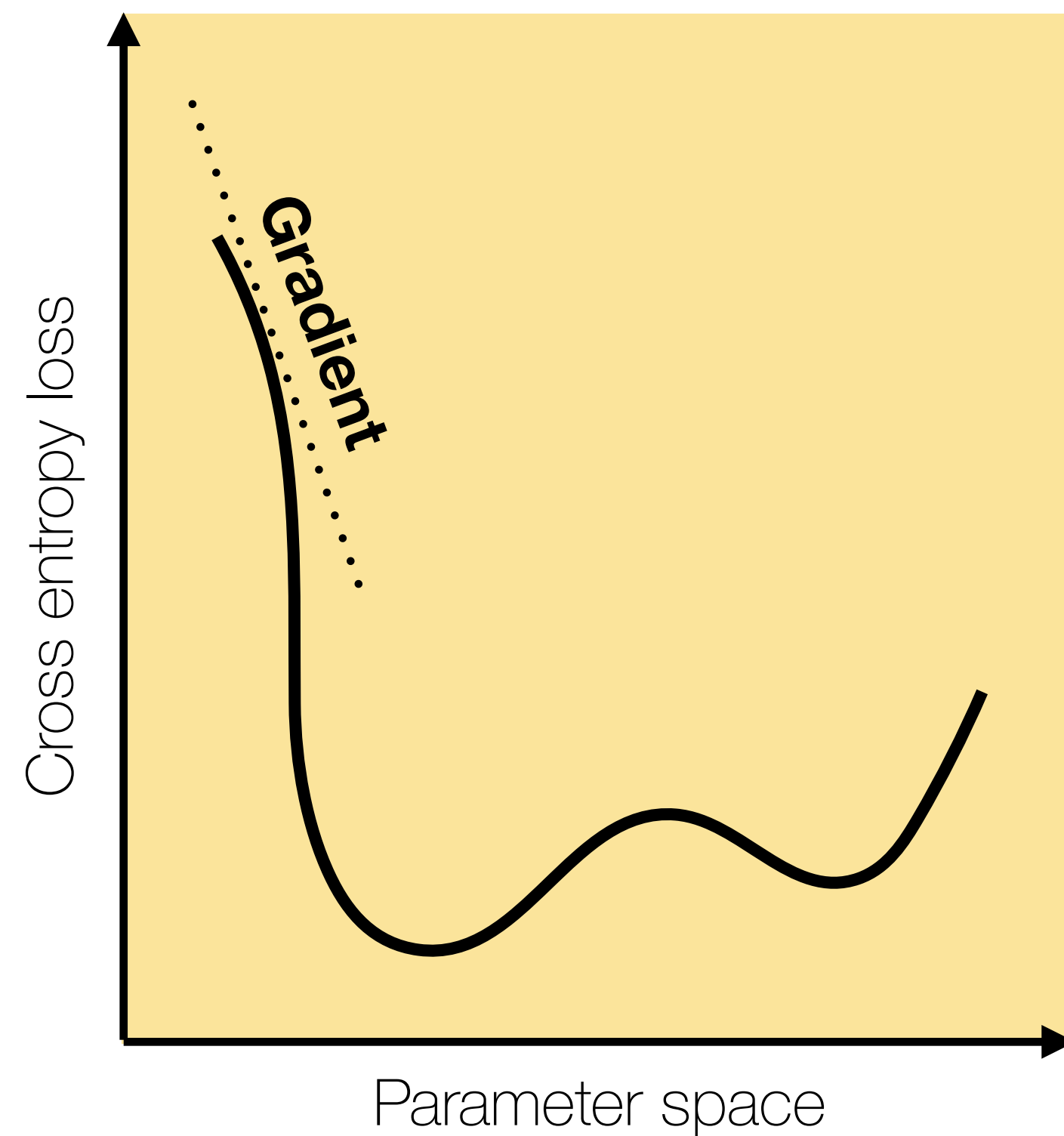
Sum over context **contexts** Sum over context **words** **Criterion**
 0 if word is absent
 1 if word is present **Model prediction**
 Softmax probability

$$-\sum_i \sum_j y_{ij} \log p(y_{ij})$$

Cross entropy loss

Error signal

and gradient descent



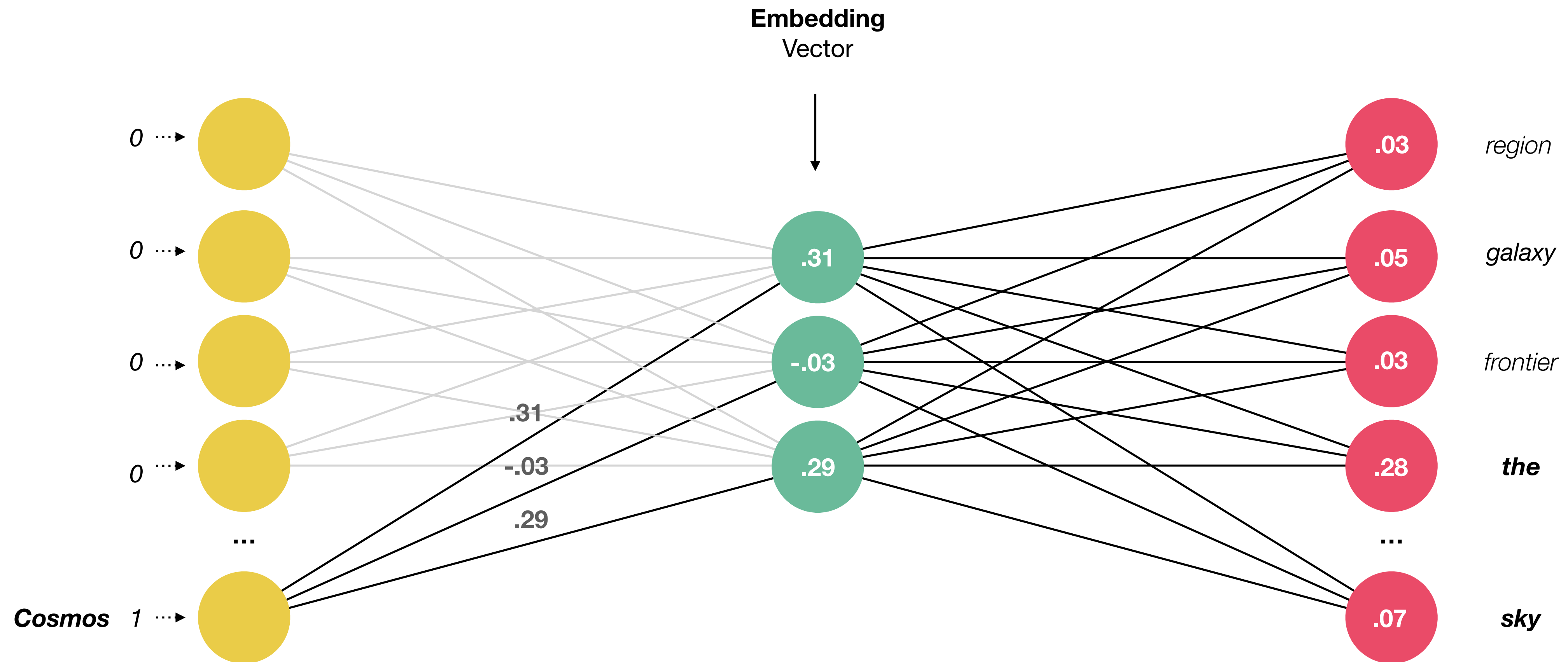
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Cross entropy loss

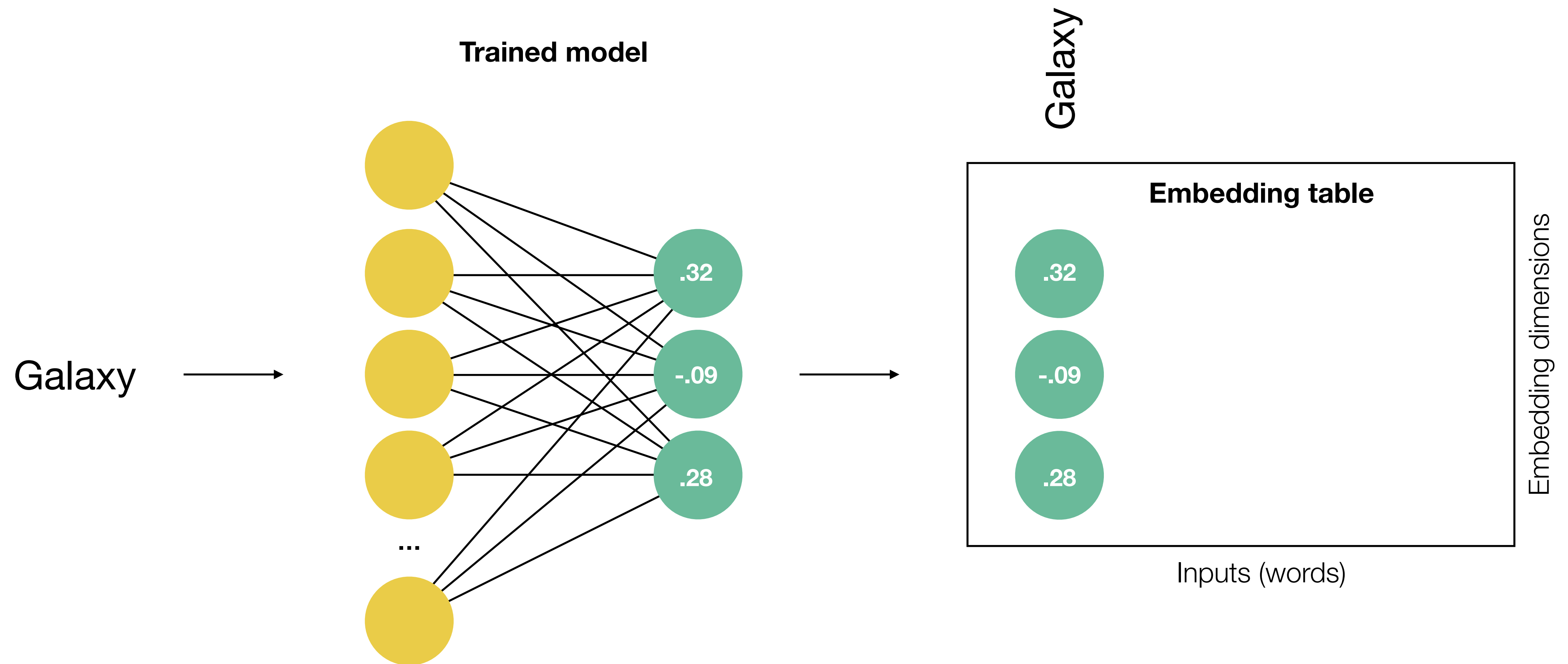
Word embedding

Embeddings = hidden activations (typically right before the output)



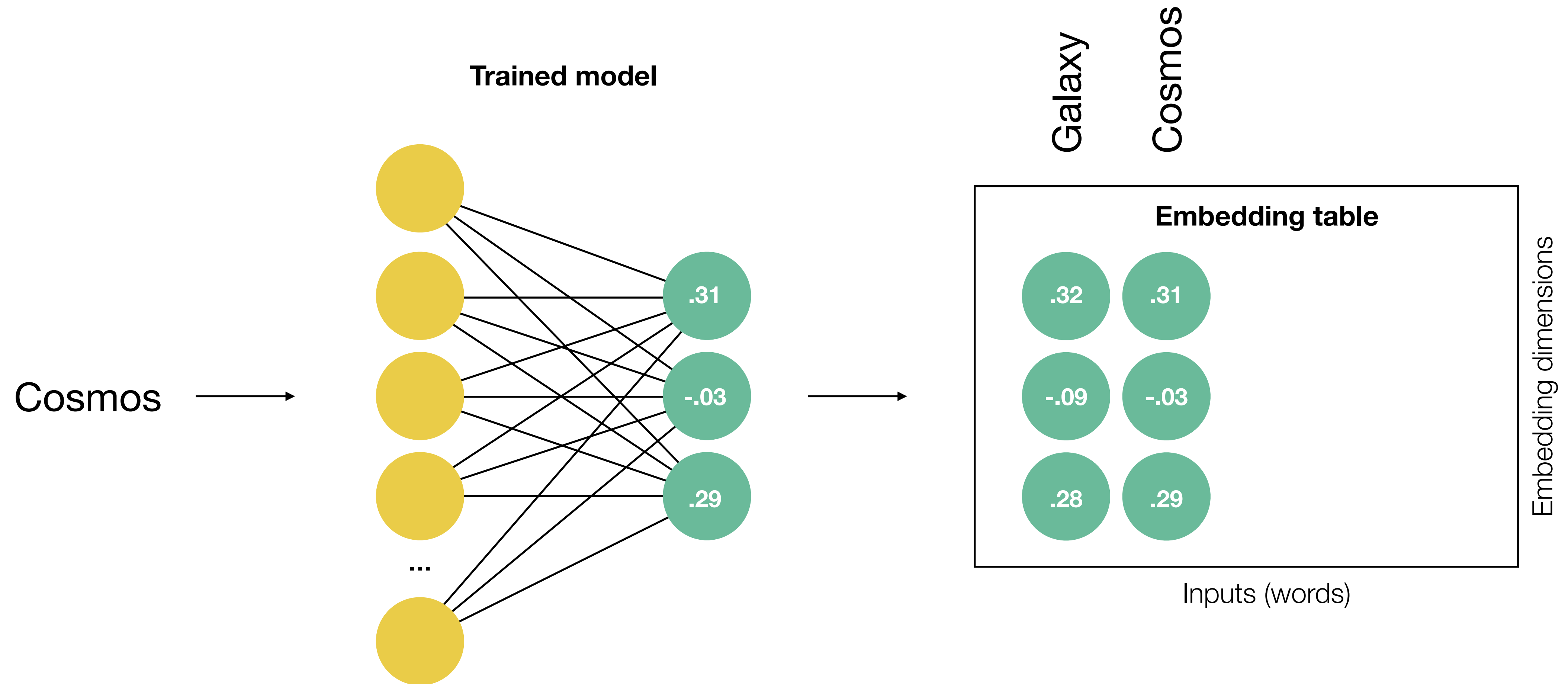
Generating embeddings

or recording hidden activations



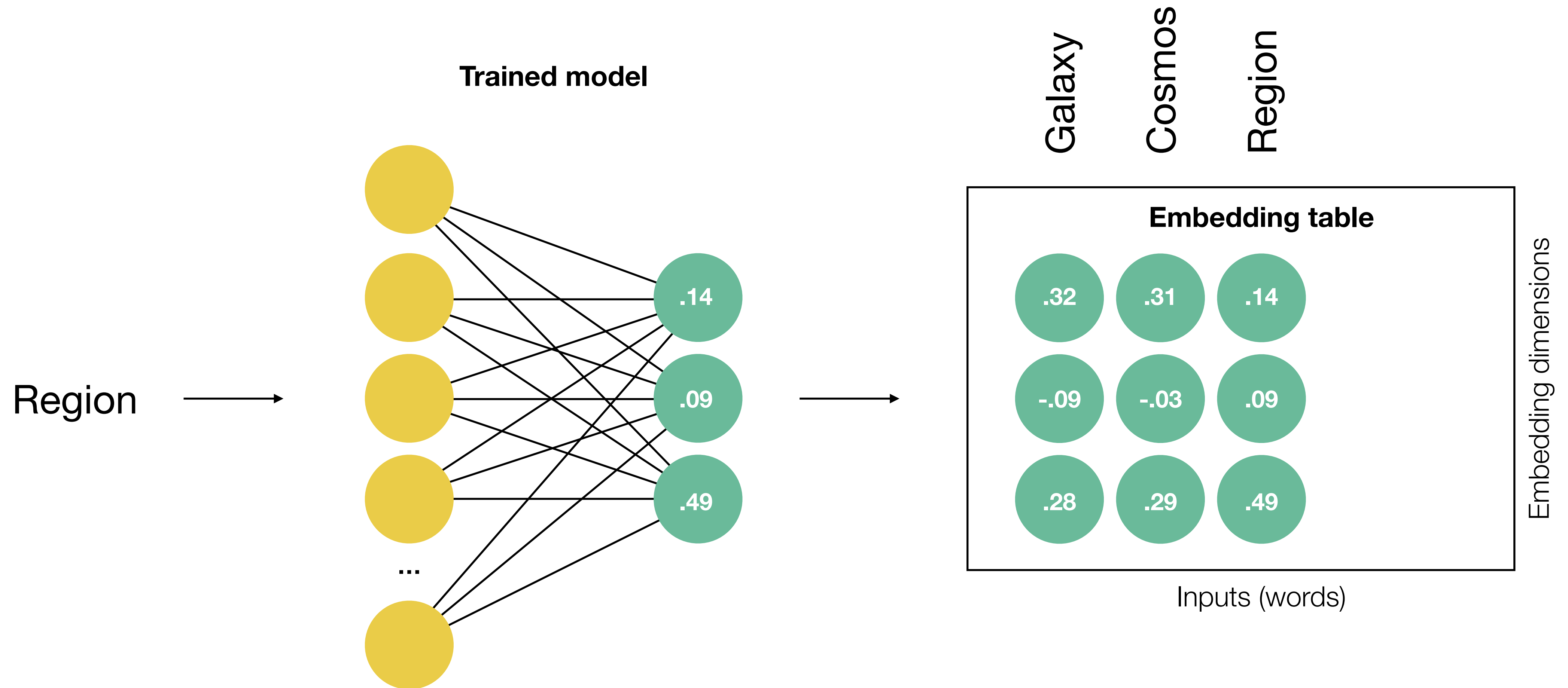
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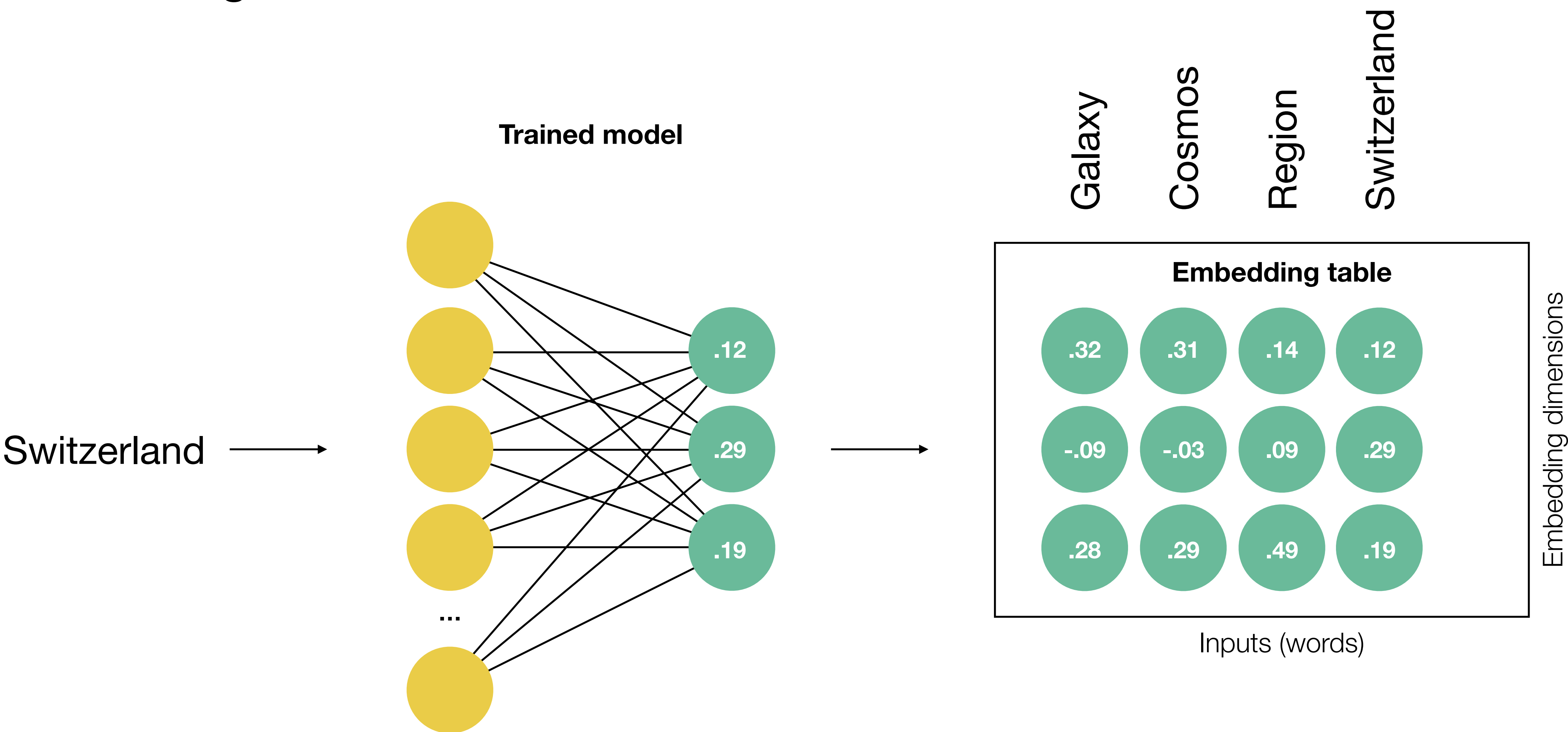
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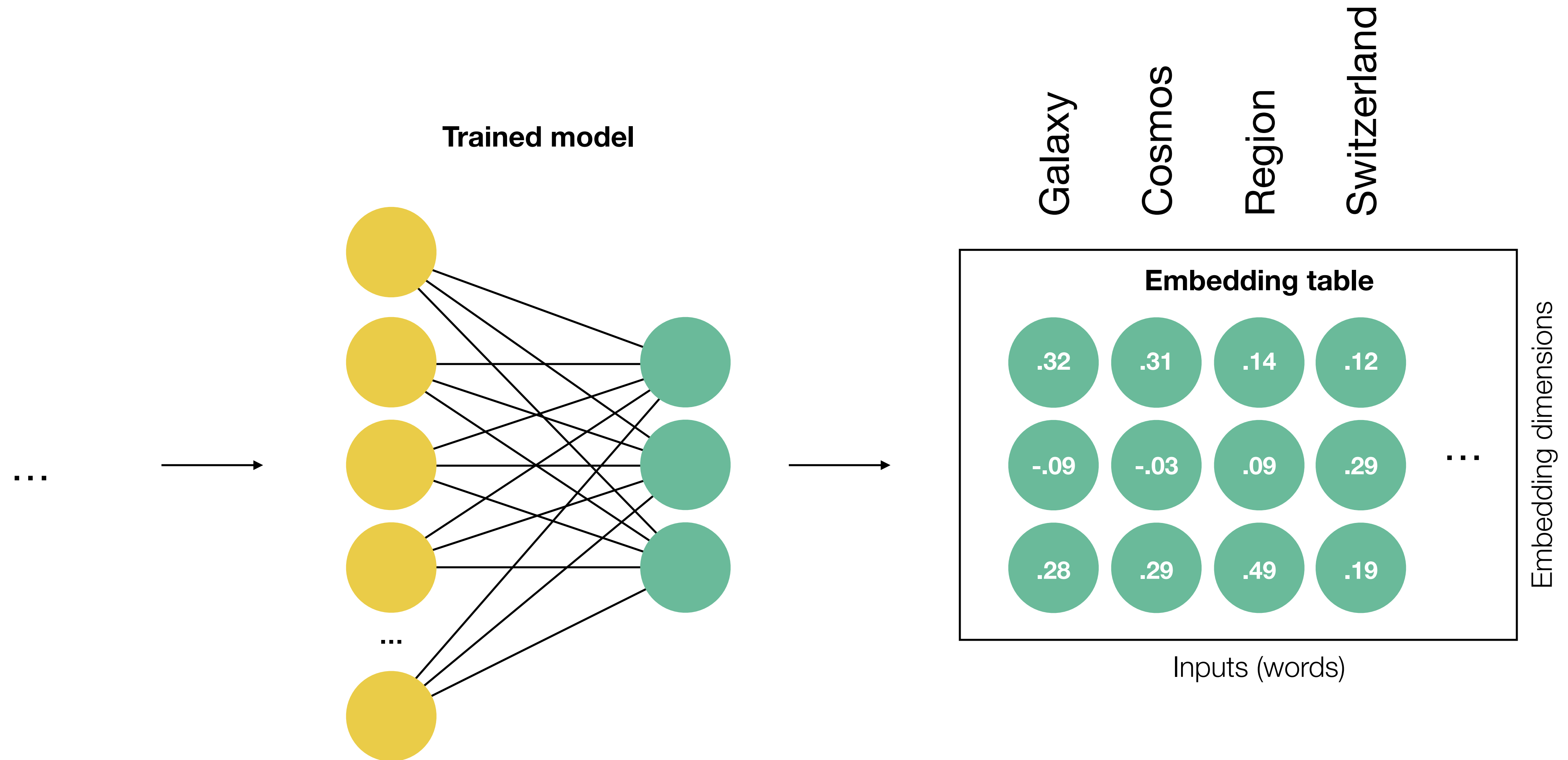
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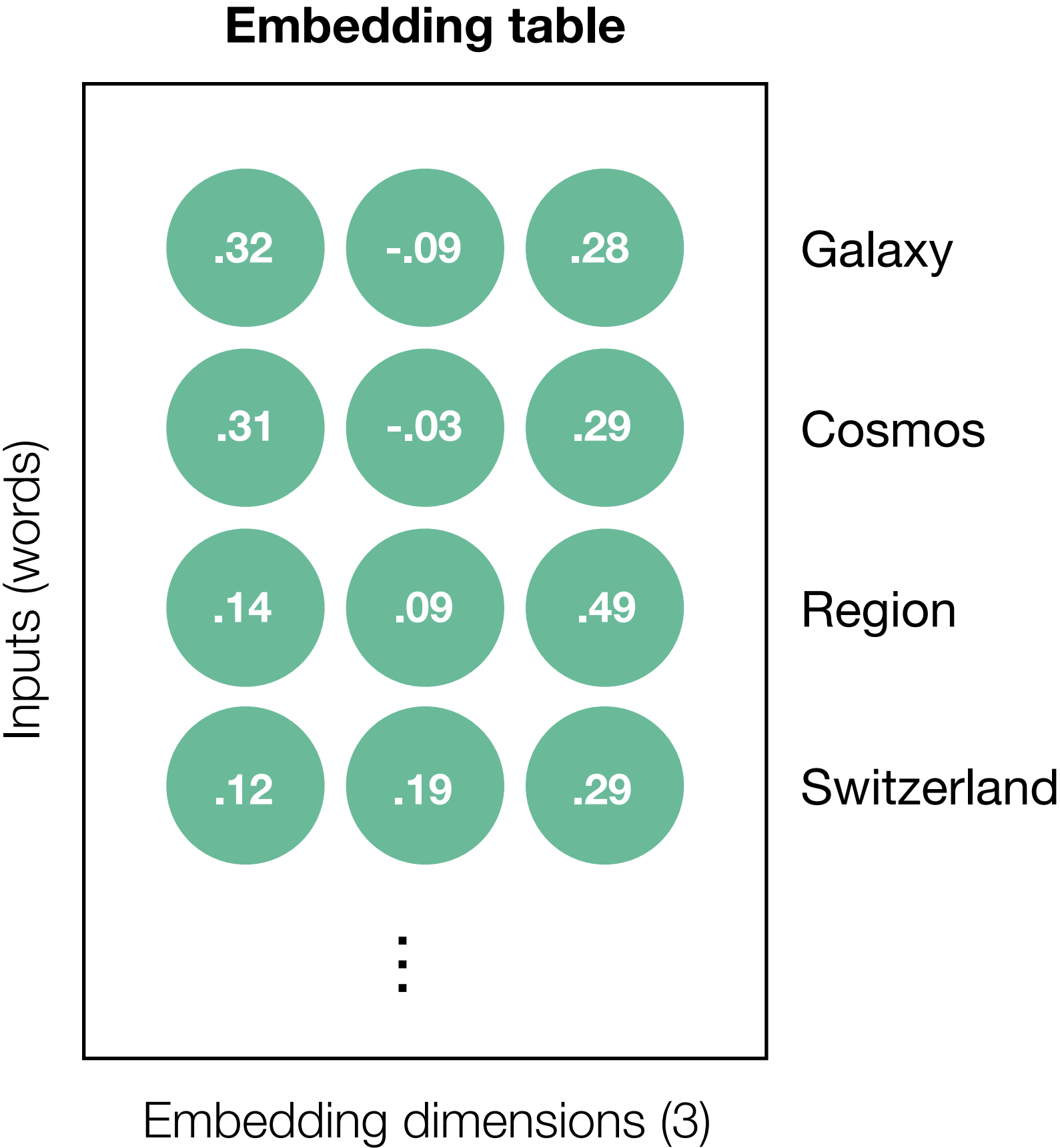
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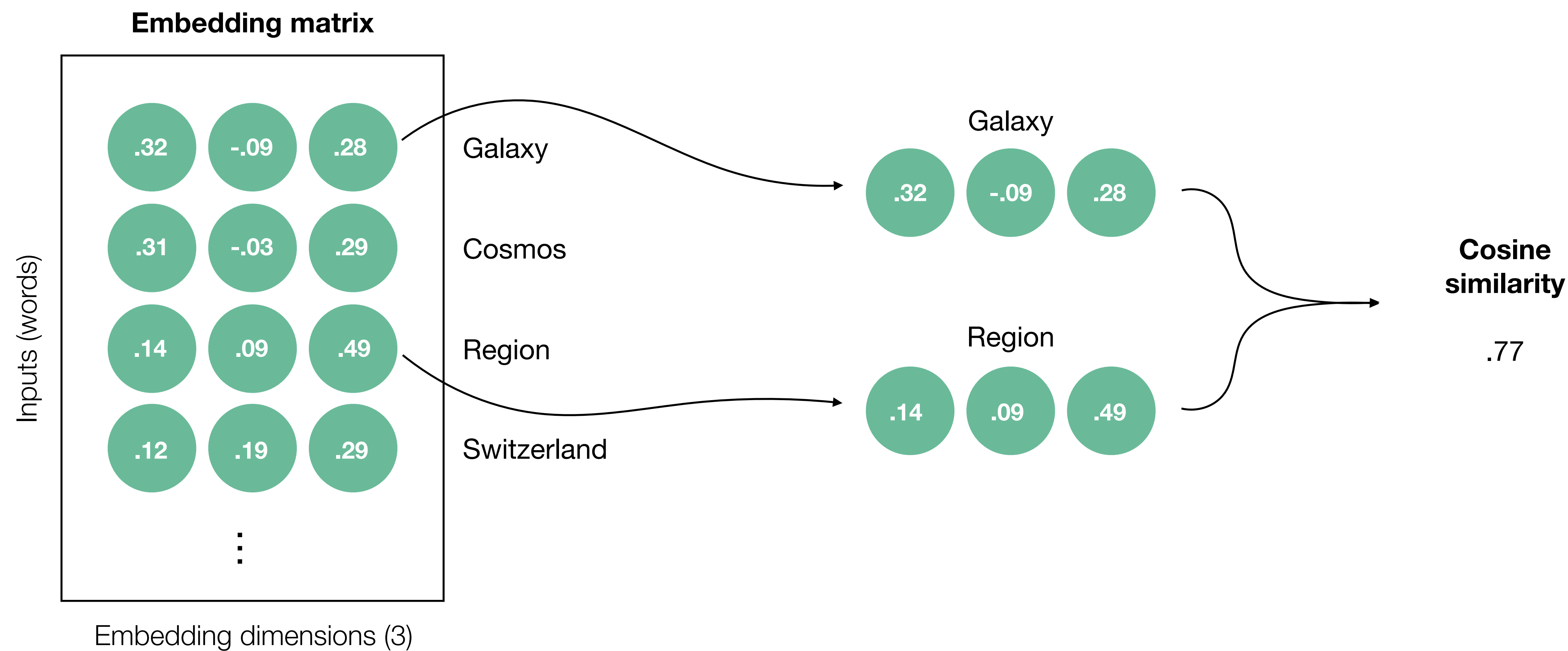
Comparing embeddings

Using cosine similarity



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